

Introduction to Economics: Core Principles and Macroeconomic Models

Comprehensive Lecture Notes – Expanded Edition

IITB Tech Community

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Contents

1	Introduction: The Foundation of Economics	2
1.1	The Concept of Scarcity	2
1.2	Microeconomics vs. Macroeconomics	2
2	The Economic Method	3
3	Economic Models: The Circular Flow	3
3.1	The Two-Sector Model	4
3.2	Expanding the Model	4
4	Foreign Capital: FDI vs. FPI	5
5	The Production Possibility Frontier (PPF)	6
5.1	Shape and Efficiency	6
6	Economic Growth and the Malthusian Trap	7
6.1	The Malthusian Theory	7
6.2	The Flaw in Malthus's Theory	7
7	Summary and Review Questions	8

1 Introduction: The Foundation of Economics

1.1 The Concept of Scarcity

The word *economics* originates from the Greek term *Oikonomia*, meaning “household management.” While historically focused on how families distributed their income, modern economics is fundamentally centered on the problem of **scarcity**: resources are limited, but human wants are effectively unlimited. Economics is therefore the study of how societies allocate scarce resources among competing, unlimited wants.

Scarcity is not simply about poverty – it is a universal condition that governs decision-making at every level of an economy, because *every* choice involves giving something up.

Key Idea

Scarcity forces choice, and choice always has a cost – this is the origin of the concept of **opportunity cost**: the value of the next-best alternative you give up when you make a decision.

Scarcity applies to every decision-maker in an economy:

- **Students:** Face a time constraint when choosing between different courses, extracurriculars, sleep, or a part-time job.
- **Households:** Face an income constraint, restricting unlimited desires for consumer goods and services.
- **Firms:** Face resource constraints (e.g., semiconductor shortages, skilled labor, capital, raw materials).
- **Governments:** Face budgetary constraints regarding infrastructure, defense, healthcare, and public spending.

Because resources are scarce, every economic system – whether a free market, a planned economy, or a mixed economy – must answer three fundamental questions:

- (1) **What** to produce (which goods and services, and in what quantities)?
- (2) **How** to produce it (what combination of labor, capital, and technology)?
- (3) **For whom** to produce it (how is output distributed among members of society)?

1.2 Microeconomics vs. Macroeconomics

- **Microeconomics:** The study of individual decision-makers, such as specific households or firms. It examines questions like: How does a rise in the price of coffee affect a café’s sales? Why does a firm hire one more worker instead of two?
- **Macroeconomics:** The study of aggregate, economy-wide phenomena, including inflation, unemployment, GDP growth, global trade, and national taxation. It examines questions like: Why did inflation rise sharply in 2022? What causes a recession?

A useful analogy: microeconomics studies individual trees, while macroeconomics studies the forest as a whole – its total size, growth rate, and health.

Practice Question

Q1. Classify each of the following as a microeconomic or macroeconomic question:

- (a) Why did Infosys's stock price rise after its quarterly earnings report?
- (b) Why did India's GDP growth slow to 4% last quarter?
- (c) Should a farmer plant wheat or sugarcane this season?
- (d) What caused the national unemployment rate to fall to 6%?

Answer: (a) Micro – concerns a single firm. (b) Macro – concerns the aggregate economy. (c) Micro – an individual producer's decision. (d) Macro – concerns the whole labor market.

2 The Economic Method

Economists analyze the world using an approach modeled on the scientific method:

1. **Observation:** Noticing an anomaly or price shift in the real world (e.g., a sudden jump in agricultural prices).
2. **Theorizing:** Formulating a hypothesis to explain the cause (e.g., poor weather restricted supply).
3. **Empirical Testing:** Analyzing historical data to verify the correlation and test the hypothesis.

Because economists usually cannot run controlled laboratory experiments on entire economies, they rely heavily on:

- **Historical and observational data** (e.g., time-series data on inflation, employment).
- **Natural experiments** – situations where a policy change or external event (like a new minimum-wage law in one state but not a neighboring one) mimics an experimental design.
- **Mathematical and statistical models** (econometrics) to test whether a relationship is statistically significant or merely coincidental.

Key Idea

Correlation is not causation. A rise in ice-cream sales and a rise in drowning deaths are correlated – both increase in summer – but ice cream does not cause drowning. Careful economic method aims to isolate true causal relationships from mere coincidence.

3 Economic Models: The Circular Flow

Because the real world is incredibly complex with countless interconnected variables, economists build simplified “model economies” using realistic assumptions. A primary example is the **Circular Flow Model**, which traces how money, goods, and resources move between different sectors of the economy.

3.1 The Two-Sector Model

In the simplest version of the economy, there are two main actors:

- **Households:** Supply the “factors of production” – Labor, Capital, and Land – to firms. In exchange, they receive income (Wages, Interest, Rent, and Profit).
- **Firms:** Use these resources to produce goods and services, which are then sold back to households in the product market.

This creates two loops flowing in opposite directions:

- A **real flow** of factors of production (from households to firms) and goods/services (from firms to households).
- A **monetary flow** of factor payments (from firms to households) and consumption expenditure (from households to firms).

In this closed, two-sector model, total income equals total expenditure equals total output – a foundational identity that underlies national income accounting.

3.2 Expanding the Model

As the model becomes more realistic, additional sectors are introduced, each creating “leakages” (money removed from the circular flow) and “injections” (money added back in):

- **Government:** Collects taxes from households and firms (a leakage). If tax rates are pushed too high, total tax revenue may actually drop due to increased tax evasion and reduced economic activity – an idea formalized in the **Laffer Curve**. The government provides public goods (infrastructure, hospitals, defense) and transfer payments in return (an injection).
- **Financial Institutions:** Banks and NBFCs (Non-Banking Financial Companies) act as intermediaries, accepting savings from households (a leakage) and channeling them as loans and investment capital to firms and consumers (an injection).
- **External (Foreign) Sector:** Introduces global trade. Imports are a leakage (money flowing out to pay for foreign goods) while exports are an injection (money flowing in). International labor migration results in remittances flowing back into the domestic economy, another injection.

Key Idea

For the circular flow to be in equilibrium: **Leakages = Injections**, i.e., Savings + Taxes + Imports = Investment + Government Spending + Exports.

Worked Example

Q: If households save more this year without any change in investment, government spending, or exports, what happens to the circular flow?

A: An increase in savings (a leakage) without a matching increase in investment (an injection) causes leakages to exceed injections. Total spending in the economy falls, which – according to the circular flow logic – tends to reduce national income and output. This is closely related to the *paradox of thrift* in Keynesian economics: if everyone tries to save more simultaneously,

aggregate demand can fall, potentially reducing overall income (and therefore total savings) rather than raising it.

Practice Question

Q2. Classify each of the following as a *leakage* or an *injection* in the circular flow model:

- (a) A household deposits part of its salary into a fixed deposit.
- (b) The government builds a new highway.
- (c) A domestic company exports textiles to Europe.
- (d) A consumer buys an imported smartphone.

Answer: (a) Leakage (saving). (b) Injection (government spending). (c) Injection (export revenue). (d) Leakage (import expenditure).

4 Foreign Capital: FDI vs. FPI

Foreign capital enters a country in two primary forms, and the distinction matters enormously for economic stability.

- **Foreign Direct Investment (FDI):** Direct investment into physical assets and operations (e.g., building a factory from scratch, or acquiring a controlling stake in an existing firm). FDI provides direct managerial control, introduces new technology and expertise, creates local jobs, and – because it is tied up in illiquid physical assets – tends to be stable during financial crises.
- **Foreign Portfolio Investment (FPI):** Passive investment in financial assets such as stocks and bonds, without any managerial control. FPI provides valuable market liquidity but is often called “hot money” because of its high volatility: investors can sell their holdings and pull funds out of a country almost instantly during a panic, which can trigger currency crises.

Feature	FDI	FPI
Control	Managerial control over the enterprise	No control; passive ownership
Liquidity	Low (illiquid, long-term)	High (can be sold quickly)
Volatility	Low, stable	High, “hot money”
Typical Horizon	Long-term (years/decades)	Short-term (can be days)
Effect on crises	Tends to stay, cushions shocks	Tends to flee, amplifies shocks

Table 1: FDI vs. FPI Comparison

Practice Question

Q3. A multinational company builds a new car manufacturing plant in India. A separate foreign hedge fund buys shares of an Indian IT company on the stock exchange. Identify which is FDI and which is FPI, and explain which is likely to be more stable during a global

financial panic, and why.

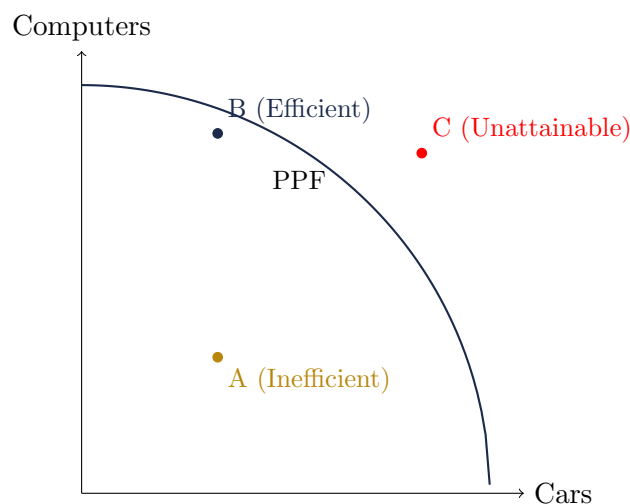
Answer: The car plant is FDI (physical, illiquid asset with managerial control). The hedge fund's share purchase is FPI (liquid financial asset, no control). FDI is more stable during a panic because physical assets like factories cannot be sold and withdrawn overnight, whereas the hedge fund can liquidate its stock position in seconds and move capital elsewhere, contributing to capital flight.

5 The Production Possibility Frontier (PPF)

The PPF is a core model used to visualize scarcity, efficiency, and trade-offs. It graphs the maximum possible output combinations of two commodities (e.g., computers and cars) given a fixed amount of resources and a fixed level of technology.

5.1 Shape and Efficiency

- **Concave (Bowed-Outward) Shape:** The PPF is bowed outward due to the **Law of Increasing Opportunity Cost**. Resources are not perfectly adaptable – a worker highly skilled at making computers may be relatively inefficient at making cars. As an economy shifts more and more resources from one industry to another, it must draw on increasingly less-suited resources, so each additional unit of one good costs more units of the other good given up.
- **Points on the Curve:** Represent points of maximum productive efficiency – all resources are fully and optimally employed.
- **Points Inside the Curve:** Represent inefficiency – unemployment, underused capital, or misallocation of resources. The economy could produce more of both goods without any new resources.
- **Points Outside the Curve:** Represent production levels that are currently *unattainable* with existing resources and technology.



Practice Question

Q4. An economy is currently producing at a point *inside* its PPF. What could cause this, and what two distinct ways could the economy move to *expand* its PPF outward (rather than just moving onto it)?

Answer: Producing inside the PPF typically reflects unemployment, underutilized factories, or inefficient allocation of resources – the economy is not using its full productive capacity. Moving *onto* the existing curve just means using existing resources more fully (e.g., reducing unemployment). To shift the entire curve *outward*, the economy needs an increase in its productive capacity itself – for example, through (1) technological advancement, or (2) an increase in the quantity/quality of resources, such as capital accumulation, a growing labor force, or improved education and skills (human capital).

6 Economic Growth and the Malthusian Trap

While points outside the PPF are currently impossible, an economy can shift its entire PPF curve outward over time – primarily through technological advancement, capital accumulation, and improvements in human capital.

6.1 The Malthusian Theory

Thomas Malthus (1798) theorized that human population grows **exponentially** (geometrically: 1, 2, 4, 8, 16...) while food production grows only **linearly** (arithmetically: 1, 2, 3, 4, 5...). Because population would eventually outstrip the food supply, Malthus predicted that humanity would inevitably face resource exhaustion, leading to mass starvation, disease, and conflict – nature’s “checks” on population growth.

6.2 The Flaw in Malthus’s Theory

Malthus failed to account for the transformative power of technology. Innovations in agriculture – fertilizers, irrigation, mechanization, and later the Green Revolution’s high-yield crop varieties – allowed food production to scale far faster than linearly, repeatedly overcoming the resource constraints Malthus predicted. In PPF terms, technological progress in agriculture has continually pushed the food-production frontier outward faster than population has grown.

Key Idea

Interestingly, Malthusian logic forms the core philosophy of popular-culture figures such as the Marvel villain Thanos, who sought to artificially and violently restrict population to preserve finite resources – a plan critics note ignores the Malthusian theory’s real-world flaw: technology and innovation, not population control, have historically been what expand a society’s resource frontier.

Practice Question

Q5. Using the PPF framework, explain in your own words why Malthus’s prediction failed to materialize on a global scale, even though world population has grown roughly eight-fold since his 1798 essay.

Answer (guideline): A strong answer should note that Malthus assumed a fixed technology (a fixed PPF), under which linear food growth against exponential population growth would indeed be unsustainable. In reality, technological change (mechanization, fertilizers, high-yield seeds, irrigation) has repeatedly shifted the agricultural production frontier outward, allowing food output to grow much faster than Malthus's linear assumption – in many periods faster than population growth itself. This illustrates why the PPF is not fixed forever: growth and innovation are themselves economic variables, not just constraints.

7 Summary and Review Questions

Key Idea

Chapter Recap:

- Economics studies the allocation of scarce resources among unlimited wants.
- Microeconomics studies individual units; macroeconomics studies the aggregate economy.
- The economic method mirrors the scientific method: observe, theorize, test.
- The circular flow model shows how households, firms, government, financial institutions, and the foreign sector interact; equilibrium requires leakages = injections.
- FDI is stable, long-term, and control-oriented; FPI is liquid, short-term, and volatile (“hot money”).
- The PPF visualizes scarcity, efficiency, and opportunity cost; its concave shape reflects increasing opportunity costs.
- Technology and resource growth shift the PPF outward over time, which is why Malthus's linear-food/exponential-population prediction did not hold.

Practice Question

Comprehensive Review Questions

1. Define scarcity and explain why it is considered the central problem of economics.
2. Give one example each of a microeconomic and a macroeconomic question not mentioned in these notes.
3. Explain, using the circular flow model, why a sudden rise in imports (with no other changes) could reduce national income.
4. Why might a government's tax revenue *fall* even after it raises tax rates? Relate your answer to the Laffer Curve idea mentioned in Section 3.2.
5. A country wants to attract stable, long-term foreign capital rather than volatile capital. Would it prioritize policies attracting FDI or FPI? Justify your answer.

6. Draw (in words) a PPF between “Healthcare Spending” and “Education Spending.” Explain what a point inside, on, and outside this curve would mean.
7. Explain why the PPF is typically drawn as concave (bowed outward) rather than as a straight line.
8. In what sense was Thanos’s plan in *Avengers: Infinity War* an application of (flawed) Malthusian logic?